Lisa Alazraki

lisa.alazraki20@imperial.ac.uk • Website • Scholar • GitHub • Linkedin • EU Citizen, UK Settled Status

I am a PhD student in machine learning and natural language processing at Imperial College London. My research aims to diagnose and improve the reasoning capabilities of large language models, with a focus on robustness, compositionality, and transferability.

EDUCATION

Imperial College London, *PhD Computing*	Supervisor: Marek Rei	*2023-2027*
Imperial College London, *MSc Computing (AI and Machine Learning)*	Classification: Distinction	*2020-2021*
The Open University, *Grad. Cert. Theoretical Statistics and Probability*	Classification: Distinction	*2019-2020*
The Open University, *BSc (Hons) Computing & IT and Mathematics*	Classification: 1st Class	*2015-2019*

EXPERIENCE

Allen Institute for AI (Ai2), *(Incoming) Research Intern* • Seattle, USA — *Jun - Sep 2026*

- Manager: Varsha Kishore. Team: Asta – AI for Science.

Meta, *Research Scientist Intern* • London, UK — *Jun - Dec 2025*

- Manager: Akhil Mathur. Team: Post-training Reasoning, Agents.
- Studied how LLM agent capabilities scale as a function of FLOPs and completion time horizon.
- Designed a marketplace-inspired adaptive multi-agent system that coordinates small agents to push the accuracy/cost Pareto frontier.

Cohere, *Research Intern* • London, UK — *Jun - Dec 2024*

- Manager: Max Bartolo. Team: Command Post-training.
- Developed a reinforcement learning pipeline for reverse engineering human preferences that boosts LLM-as-a-judge evaluation.
- Investigated implicit learning from mistakes, showing LLMs attain higher accuracy when not shown explicit corrective feedback.
- Completed two distinct research projects at the same time, resulting in first-author papers at NeurIPS and EMNLP respectively.

Google, *Research Intern* • Amsterdam, Netherlands — *Jun - Sep 2023*

- Manager: Thomas Mensink. Team: Perception.
- Developed a model-ensembling framework for knowledge-intensive VQA that beats SOTA by 5% on Encyclopedic-VQA.
- Presented the resulting publication at ICBINB at NeurIPS 2023.

Google, *Student Researcher* • London, UK — *Oct - Dec 2022*
Research Intern • Zurich, Switzerland — *Jun - Sep 2022*

- Manager: Hamza Harkous. Team: Applied Privacy Research.
- Developed a new pipeline for retrieval-augmented generation of user issues that was deployed to production.
- Improved recall of existing issues by 10× over the previous model, with comparable semantic accuracy for new issue generation.
- Granted a global patent as co-inventor of the overall system for navigating user feedback.

SELECTED PAPERS

Scaling Small Agents Through Strategy Auctions, *ICML 2026* — *2026*
L. Alazraki, W. F. Shen, Y. Bachrach, A. Mathur

Rethinking Rubric Generation for Improving LLM Judge and Reward Modeling for Open-ended Tasks, *in review* — *2026*
W. F. Shen, X. Qiu, C. Whitehouse, L. Alazraki, S. Goel, F. Barbieri, T. Willi, A. Mathur, I. Leontiadis

AgentCoMa: A Compositional Benchmark Mixing Commonsense and Mathematical Reasoning, *ACL 2026* — *2025*
L. Alazraki, L. Chen, A. Brassard, J. Stacey, H. A. Rahmani, M. Rei

Improving the OOD Performance of Closed-Source LLMs Through Strategic Data Selection, *EACL 2026 Findings* — *2025*
J. Stacey, L. Alazraki, A. Ubhi, B. Ermis, A. Mueller, M. Rei

Reverse Engineering Human Preferences with Reinforcement Learning, *NeurIPS 2025 (Spotlight)* — *2025*
L. Alazraki, T. Yi-Chern, J. A. Campos, M. Mozes, M. Rei, M. Bartolo

No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context, *EMNLP 2025 (Oral)* — *2025*
L. Alazraki, M. Mozes, J. A. Campos, T. Yi-Chern, M. Rei, M. Bartolo

Enhancing LLM Robustness to Perturbed Instructions: An Empirical Study, *ICLR 2025 BuildingTrust* — *2025*
A. Agrawal*, L. Alazraki*, S. Honarvar, M. Rei (*Equal contribution)

How can representation dimension dominate structurally pruned LLMs?, *ICLR 2025 SLLM* — *2025*
M. Xu, L. Alazraki, D. Mandic

Meta-reasoning Improves Tool Use in Large Language Models, *NAACL 2025 Findings* — *2024*
L. Alazraki, M. Rei

How (not) to ensemble LVLMs for VQA, *NeurIPS 2023 ICBINB* — *2023*
L. Alazraki, L. Castrejon, M. Dehghani, F. Huot, J. Uijlings, T. Mensink

AWARDS

Placement Award, *Alan Turing Institute* *2024*
Selected as one of ∼30 resident PhD students for the Institute's Enrichment Scheme 2024/25.

ICC Poster Competition First Prize, *Imperial College London* *2024*
Awarded for best poster presentation at the Imperial Computing Conference.

IET Postgraduate Prize, *Institution for Engineering and Technology* *2024*
Awarded for 'excellence in postgraduate research' at the IET Postgraduate Research Awards.

Best Student Paper Award, *Institute of Electrical and Electronics Engineers* *2023*
Awarded at the IEEE International Conference on Cognitive Machine Intelligence.

Distinguished Dissertation Award, *Imperial College London* *2021*
Awarded for 'outstanding MSc dissertation in terms of technical achievement and presentation'.

Faculty Commendation, *The Open University* *2017*
Awarded for achieving one of the highest results out of ∼500 students in Algorithms, Data Structures and Computability.

Leslie Walshaw Award, *The Open University* *2016*
Awarded for achieving the highest overall grade of all students in the Greater London region in mathematics module MST125.

LEADERSHIP AND COMMUNITY

Supervision	Conceived 10+ MSc/MEng thesis specifications and co-supervised 15 students across machine learning and NLP
Mentorship	Hold regular office hours advising applicants on graduate admissions, particularly from non-traditional backgrounds
Organisation	Co-organise a weekly NLP reading group at Imperial College London, hosting both internal and external speakers
Teaching	Graduate teaching assistant across five courses spanning computer science, machine learning, deep learning, and NLP
Reviewing	Reviewer for ACL Rolling Review, COLING, COLM, ICML, and NeurIPS, as well as associated workshops

SELECTED TALKS

Invited

Pattern Matching vs. Procedural Reasoning in Language Models, *CaMLSys, University of Cambridge* *2026*
Reverse Engineering Human Preferences with Reinforcement Learning, *Lancaster AI, Lancaster University* *2026*
Reverse Engineering Human Preferences with Reinforcement Learning, *Breaking Topics in AI Conference, I-X Imperial* *2025*
Benchmarking LLMs on Mixed-type Compositional Reasoning, *Open Science Community Lightning Talks, Cohere Labs* *2025*

Contributed

Mixing Commonsense and Mathematical Reasoning in Real-World Scenarios, *NLP Special Interest Group, Alan Turing Institute* *2026*
Pattern Matching vs. Procedural Reasoning in Language Models, *Imperial NLP, Imperial College London* *2026*
Plan Auctions for Task-Aware Selection of LLM Agents Across Scales, *Meta Superintelligence Labs Tech Talks, Meta* *2025*
No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context, *Conference on Empirical Methods in NLP* *2025*
Overcoming the Limitations of Autoregressive Generation, *Imperial NLP, Imperial College London* *2024*
A Retrieval-Augmented Generation Pipeline for User Issue Classification, *T5 and Beyond Monthly Talks, Google* *2022*

GRANTS AND SCHOLARSHIPS

IX-WAI Early Career Development Grant, *I-X Imperial* *2025*
OpenAI API Researcher Access Grant, *OpenAI* *2025*
Annual Grant, *Sir Richard Stapley Trust* *2024*
General Fund Grant, *Imperial College Trust* *2023*
IET International Travel Award, *Institution for Engineering and Technology* *2023*
Annual Grant, *Sir Richard Stapley Trust* *2023*
EPSRC Doctoral Scholarship, *Imperial College London* *2022*
DeepMind MSc Scholarship, *Imperial College London* *2020*

PATENTS

Deep Learning System for Navigating Feedback, *US Patent US20250272485A1, CN Patent CN119053957A, WO Patent WO2023224672A1*
Inventors: H. Harkous, S. T. Peddinti, R. Khandelwal, A. Srivastava, N. Taft, L. Alazraki. Assignee: Google LLC

TRAINING AND SKILLS

Summer schools attended	Oxford Machine Learning Summer School MLx Representation Learning & GenAI 2025 + MLx Health & Bio 2025
Programming languages	Python, TypeScript, JavaScript, Java, Lua, MATLAB/Octave, Maxima, C++, Solidity, Prolog, Unix/Bash, HTML, CSS
Libraries / frameworks	PyTorch, TensorFlow, Transformers, vLLM, TRL, NumPy, Pandas, Scikit-learn, Jinja2, Matplotlib, Plotly, WanDB